

How various predictors (age, gender, bmi..etc) effect claim amounts

Appendix

1. Initial data exploration

1.1

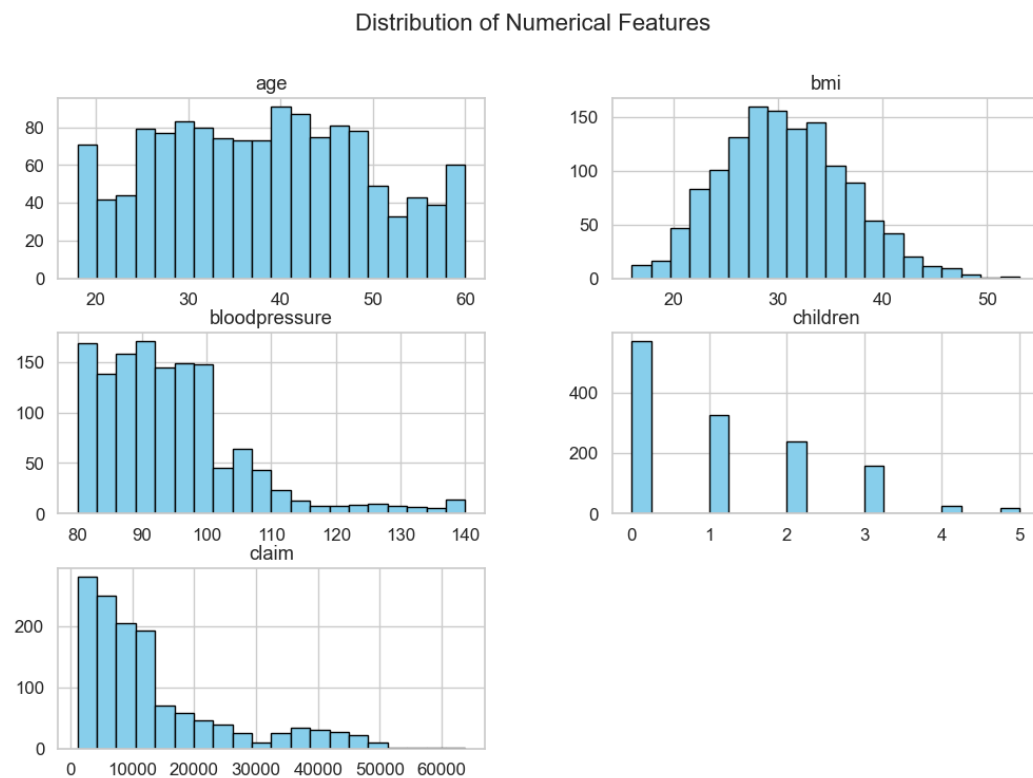
Check for rows with empty variable numbers.

Since out of 1340 rows, only 8 rows have missing numbers, it was decided that those rows with missing values to be dropped.

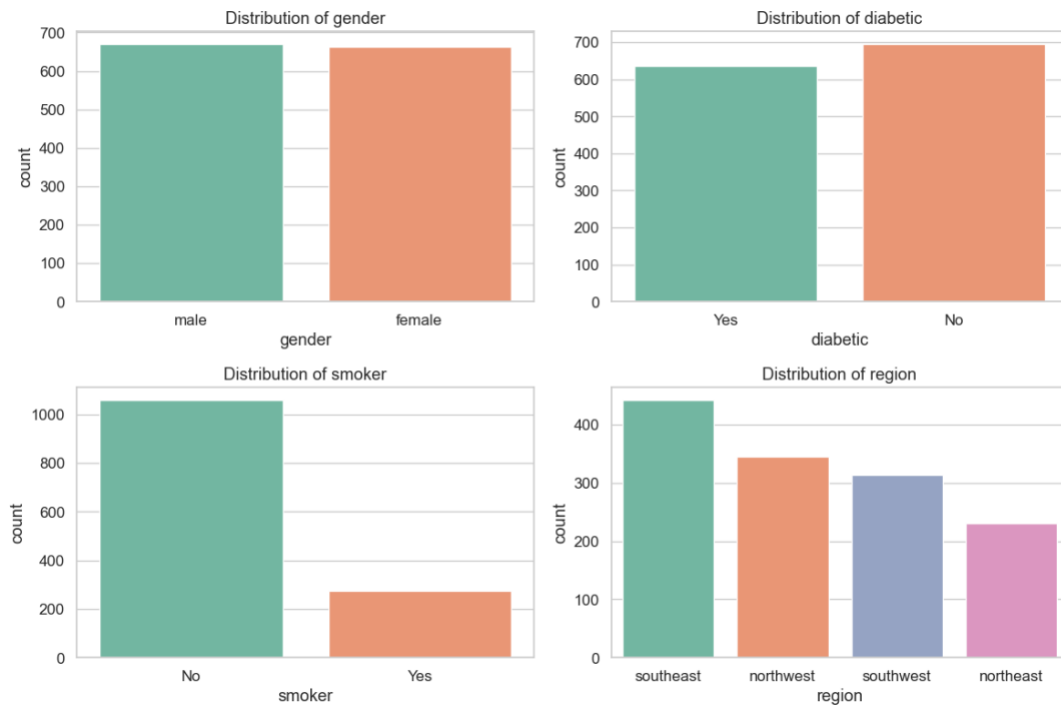
1.2

We have both numeric and categorical predictors.

First we visualise the distribution of the numerical variables (using histograms):

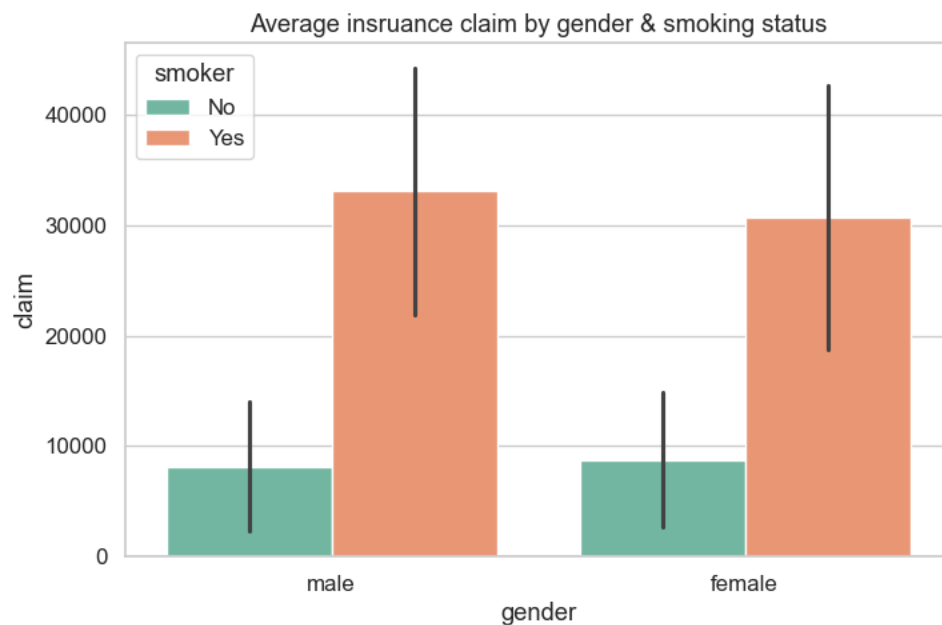


Then we visualise the distribution of categorical variables (using bar plots):



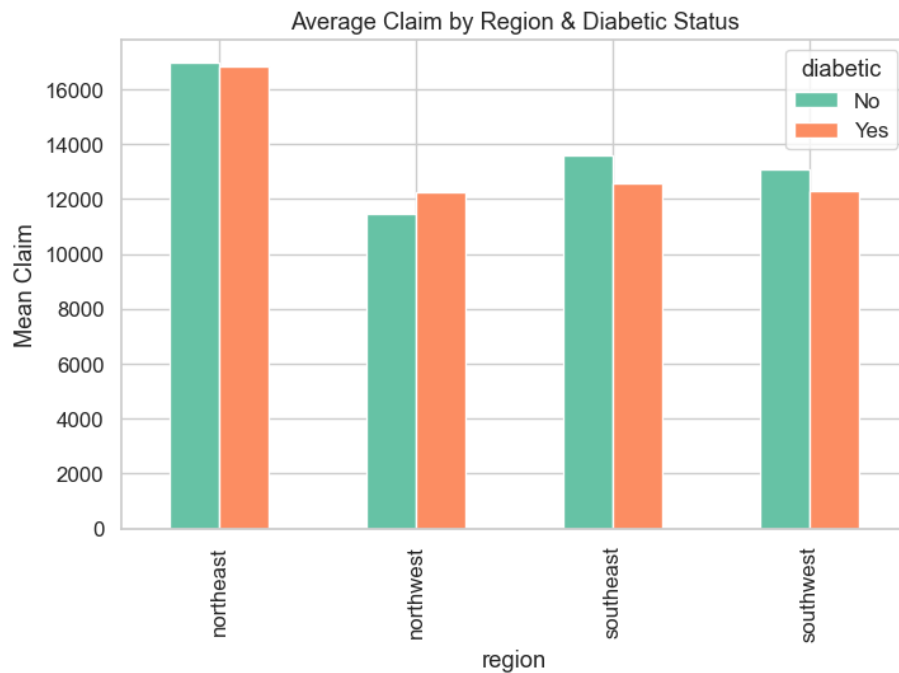
1.3 (Two way summaries of the relationship between potential predictors and target variables – for categorical variables as can't find correlations for them)

Relations between gender & smoking status AND “average” insurance claims



- From this, whether a person is male or female does not affect the claim amount
- However, being a smoker does increase the average claim amount

Relations between region & diabetic status AND average insurance claims



- Generally northeast has higher claims
- Whether diabetic or not doesn't really lead to higher mean claims

Relations between region & smoker AND average insurance claims

smoker	No	Yes
region		
northeast	11666.112195	29673.536269
northwest	8076.203415	30192.002759
southeast	7444.144872	34844.997253
southwest	8294.754102	32269.064138

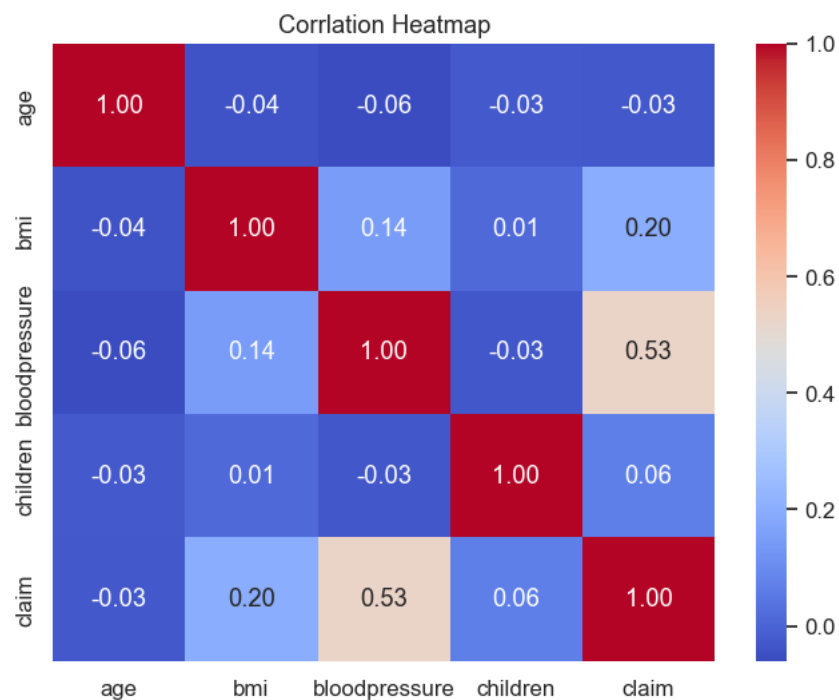
- Smokers have higher mean claims (the regions doesn't matter)
- For non-smoking individuals, northeast has noticeable higher mean claims

Relations between diabetic & children AND mean insurance claims

diabetic	No	Yes
children		
0	12967.395398	11985.289857
1	12730.455810	12732.055724
2	15567.767583	14579.360417
3	13807.612892	17091.258649
4	14106.630000	13573.352500
5	8519.043636	9205.594286

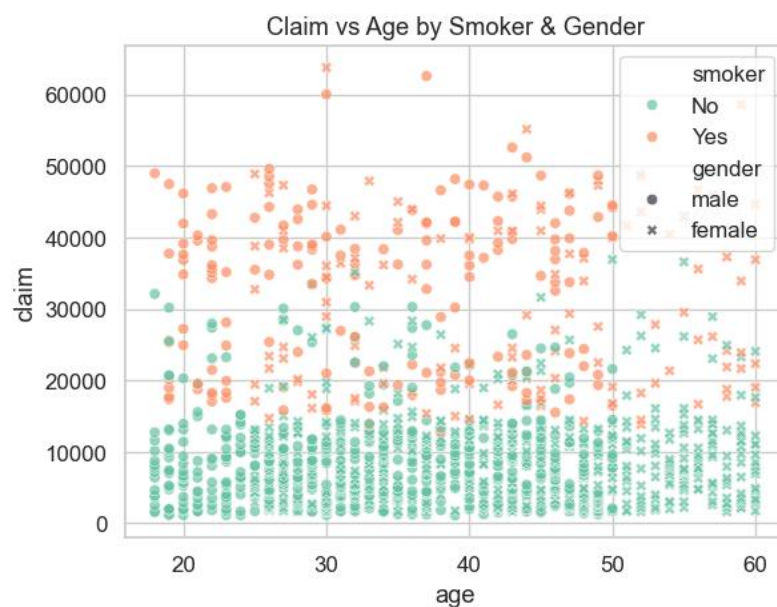
- No noticeable trends based in diabetic & children

1.4 correlation between predictors (ofc numerical variables only)

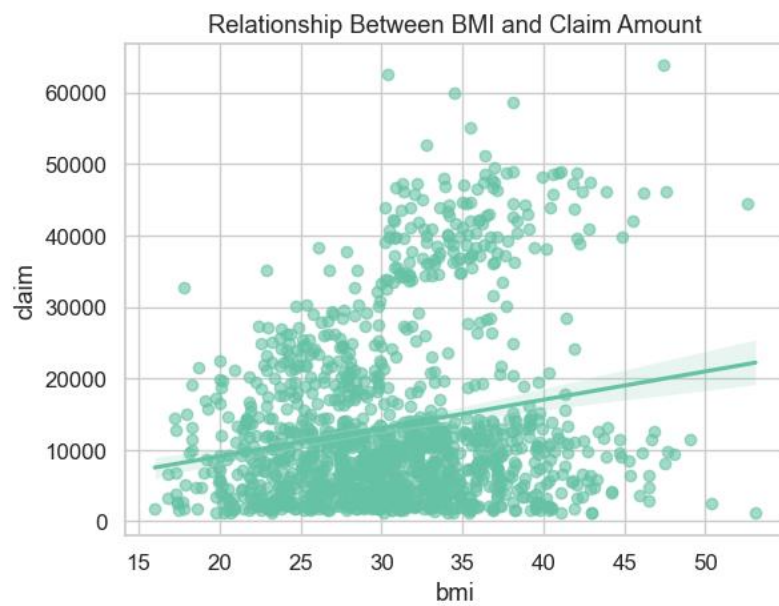


- No noticeable correlation besides blood pressure and claim

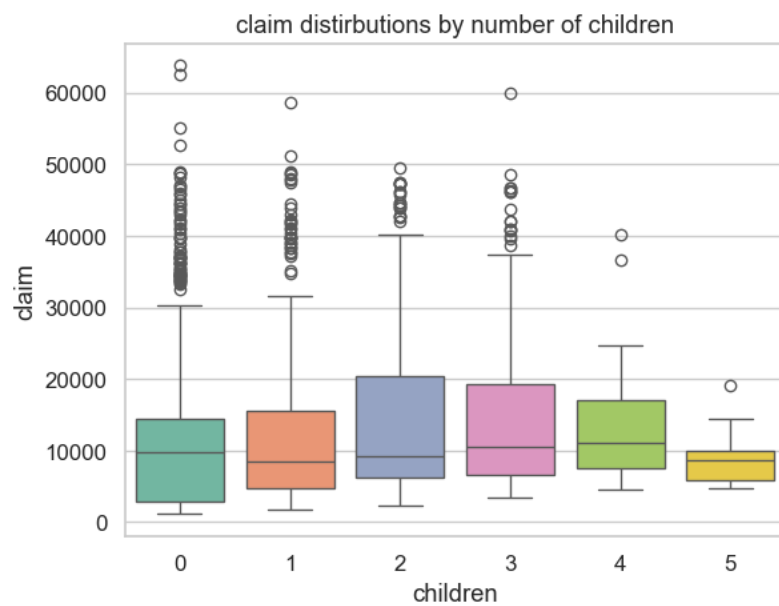
1.5 relationship analysis (1/2/3-way summaries)



- Clearly smokers have higher claims
- gender and age seem to have no effect

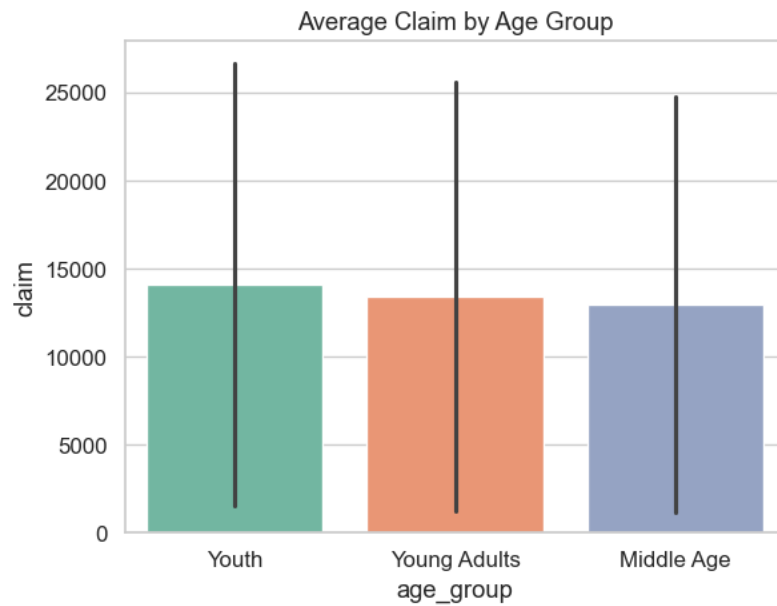


- bmi: 30~45 seems to show more frequent higher claim amounts

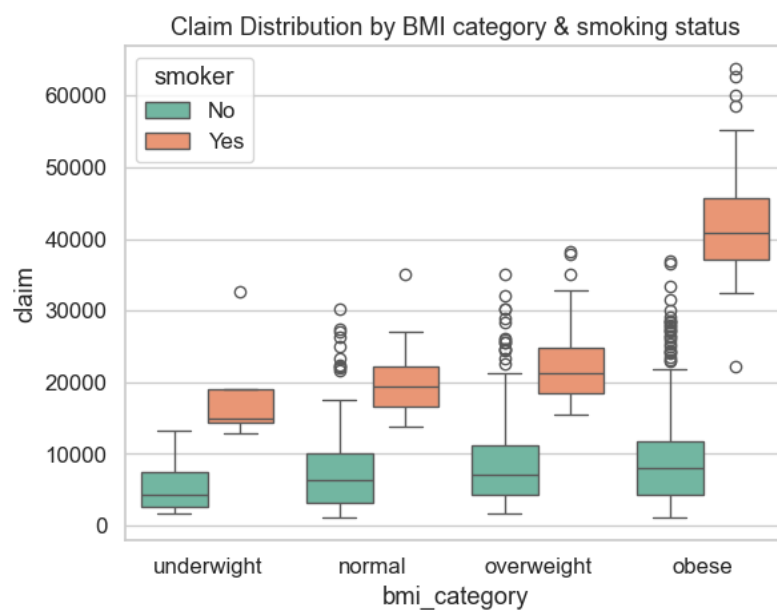


- individuals with 2 and 3 number of children tend to have Higher claims

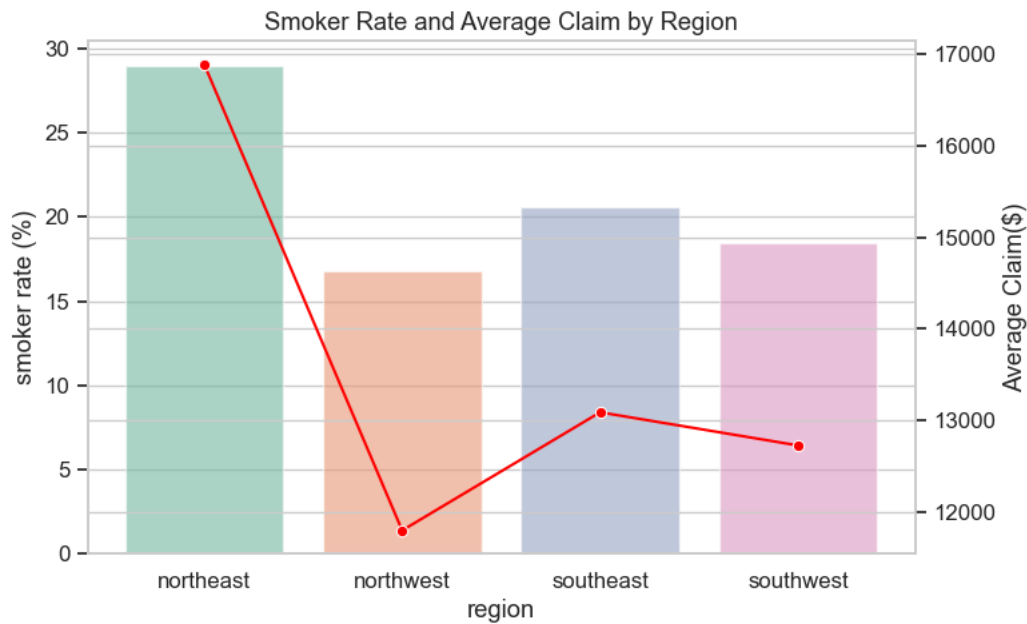
1.6 creating new variables (going from continuous to band to see group effects)



- for age, group them into diff categories
- based on age group, no significant differences (just a slight decrease in claims on older group)



- for bmi, grouped them into diff categories
- also hue the smoker
- clearly, smokers have higher claims
- also, the higher bmi groups tend to have higher claims (especially for smoking and obese)



- for each region, calculated two variables: smoker rate (barplot) and average claim (line)
- from this can see that the region with higher smoking rates had higher claims

2. Linear Models

2.05

70% of the data was used as a training data to train the models represented below.

Using the other 30% as the test data, the MAE and RMSE for each trained models were calculated.

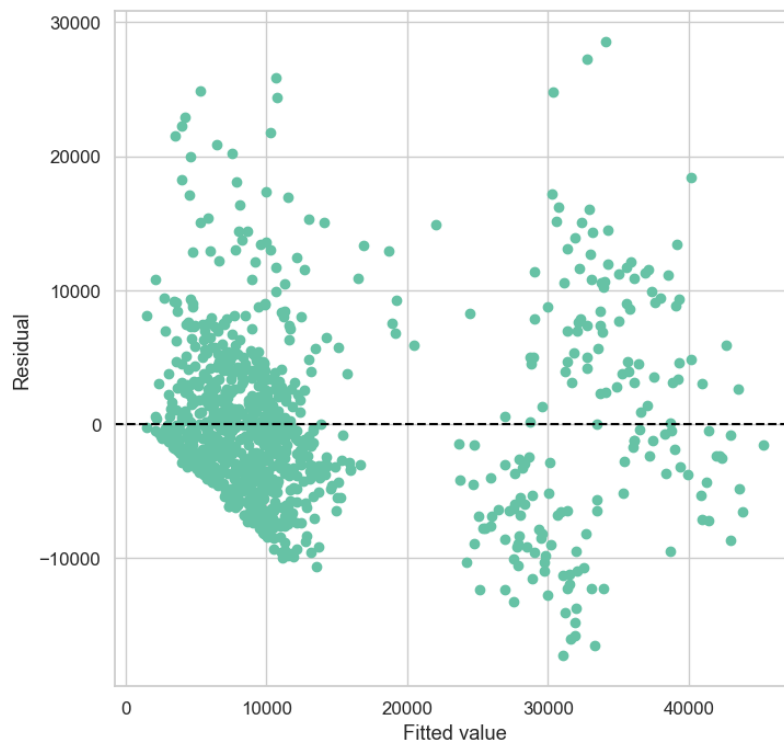
2.1 Multiple linear regression

	coef	std err	t	P> t
intercept	-16820.000000	2917.523000	-5.764000	0.000000
gender[male]	-210.332700	474.009000	-0.444000	0.657000
bloodpressure	226.032900	21.418000	10.553000	0.000000
diabetic[Yes]	-416.367200	433.263000	-0.961000	0.337000
children	636.601600	176.320000	3.610000	0.000000
smoker[Yes]	21490.000000	608.284000	35.336000	0.000000
region[northwest]	-1531.970000	674.889000	-2.270000	0.023000
region[southeast]	-2131.266500	639.566000	-3.332000	0.001000
region[southwest]	-1345.416400	696.712000	-1.931000	0.054000
age_group[Young Adults]	1304.637300	728.212000	1.792000	0.074000
age_group[Middle Age]	1238.796500	811.980000	1.526000	0.127000
bmi_category[normal]	1730.483800	1974.912000	0.876000	0.381000
bmi_category[overweight]	2786.298500	1944.492000	1.433000	0.152000
bmi_category[obese]	6450.012200	1931.433000	3.339000	0.001000

Response variable as claim and gender, bloodpressure, diabetic, children, smoker, region, age_group, bmi_category as predictor variables.

- It seems like gender, diabetic, age-group, bmi_category[normal], bmi_category[overweight] have no impact on claim size

2.1.1 fitted vs residuals

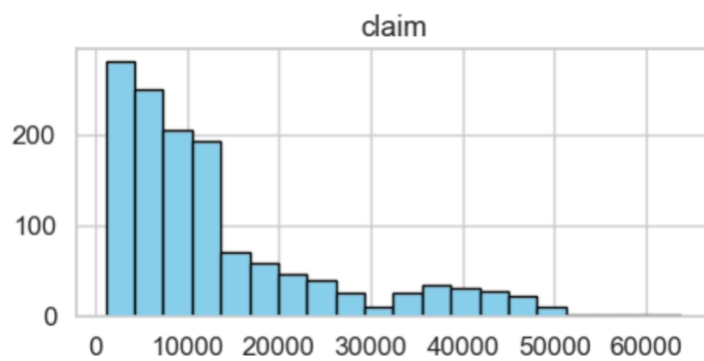


- With fitted value, the spread of residual changes -> non-constant variance of the error term
- Residuals show systematic patterns, within the 1000~15000 fitted values, where the spread of residuals increase by more negative residuals present as fitted values increase → non-linearity
- Hence, “linear” regression is not a valid fit

2.2 GAMs

Since linear regression wasn't a valid fit, let us consider GAMs with splines to capture non-linearities

2.2.1 Gamma GAM



Recall that claims are always positive.

Also, the right skewed long tail means distribution is not symmetric / normal.

Hence, Gamma GAM with log link function would be appropriate.

The GAM was constructed as follows:

- A smoother spline effect on bloodpresssure variable, using 10 spline basis functions
- A smoother spline effect on children variable , using 10 spline basis functions
- A categorical effect for all other categorical variables with no smoothing penalty done

The result:

```

GammaGAM
=====
Distribution:          GammaDist Effective DoF:          22.3963
Link Function:        LogLink Log Likelihood:          -9436.8574
Number of Samples:    932 AIC:          18920.5074
                      AICc:          18921.7652
                      GCV:           0.388
                      Scale:          0.6308
                      Pseudo R-Squared: 0.5408
=====
Feature Function      Lambda      Rank      EDoF      P > x      Sig. Code
=====
s(0)                  [0.6]      10       6.8      4.97e-11    ***
s(1)                  [0.6]      10       4.8      4.90e-03    **
f(2)                  [0.6]       2       1.0      5.53e-01
f(3)                  [0.6]       2       1.0      7.97e-01
f(4)                  [0.6]       2       1.0      1.11e-16    ***
f(5)                  [0.6]       4       2.9      4.98e-04    ***
f(6)                  [0.6]       3       1.9      4.10e-01
f(7)                  [0.6]       4       2.9      1.53e-05    ***
intercept             [0.6]       1       0.0      1.11e-16    ***
=====
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

3. Tree-Based Models

Tree-based models were also considered to model this data.

3.1 Bagging

First, the default bagging method (considering all predictor variables, when doing the split in the tree) was used to do regression by using the training data. The test MSE turned out to be 313,102,25.

3.2 Random Forests

Next, random forests modelling method was used (which considers only 4 predictor variables out of 13 variables when doing split in the tree). The MSE was slightly lower with 310,443,41, indicating that random forests method was slightly better.

3.2.1 The importance of each variable

Extracting the `feature_importances_values` from the fitted random forests model, the importance of each variable was viewed:

	importance
smoker[Yes]	0.516534
bloodpressure	0.262936
bmi_category[obese]	0.062218
children	0.037925
bmi_category[normal]	0.017742
bmi_category[overweight]	0.016934
gender[male]	0.015891
diabetic[Yes]	0.015109
region[southeast]	0.012576
age_group[Young Adults]	0.011670
region[southwest]	0.011138
region[northwest]	0.009963
age_group[Middle Age]	0.009364

- Aligning with previous findings, smoker variable was still regarded as one of the most important variables.

3.3 Boosting

Finally, boosting (specifically XGBoost) was considered with 3000 trees of max depth 3 and learning rate λ of 0.01.

The test MSE obtained after training the model was 30339613, lower than bagging and random forests.

4.1 Results from modelling

	MAE	RMSE
XGBoost	3889.091467	5508.140633
Random Forest	3947.737594	5571.744937
Bagging	4014.835540	5595.554107
Linear Regression (OLS)	5015.581800	6633.237627
Gamma GAM (pyGAM)	4709.963626	6640.357906

From the results in the table, XGBoost achieved the lowest MAE and also the lowest RMSE, meaning it is best on both typical dollar error and big-claim error control respectively. Random forests have slightly worse MAE/RMSE than GBM, often more robust with less tuning. The OLS / Gamma GAM underfit the real nonlinear structure, indicated via higher MAE/RMSE.

Ultimately, using MAE as the primary severity metric, XGBoost provided the best predictive performance and is selected as the preferred claim severity model.

5.1 Deployment

The model was deployed as a Streamlit web app. User inputs are transformed using the same saved feature engineering pipeline and category encodings as the training data,

then passed to the stored trained model to produce real time insurance payment predictions.

